Comparative Analysis of Unsupervised Clustering Algorithms on Housing Data

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***Abstract*—No one can deny that unsupervised learning has an important role in data exploration and pattern discovery, specifically when labeled data is unavailable.In this paper, we compare and analyze many clustering algorithms that have been applied to a housing dataset. This dataset contains socioeconomic and demographic features. We did this comparative and evaluative study to see the different performance of unsupervised algorithms on the same dataset. We used for clustering algorithms four unsupervised algorithms, which are: K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models (GMM).Before we start training the models, we pre-processed the dataset through data cleaning, standardization, and outlier treatment using the Interquartile Range (IQR) method. Principal Component Analysis (PCA) was used for two-dimensional visualization of the generated clusters.**

The result of this work showed that K-Means and GMM created more structured and balanced clusters, during Agglomerative Clustering obtained hierarchical group separation. DBSCAN yielded strong capabilities in detecting noise and outliers but showed sensitivity to parameter selection and generated fewer clearly separated clusters on this dataset. to Visual analysis we used PCA demonstrated differences in cluster compactness and separation between the models.

This comparative study highlighte that clustering performance strongly depends on data distribution and algorithm characteristics. The results provide valuable Visions into the points of power and points of weak of different clustering wayes for exploratory data analysis

***Keywords*—Unsupervised Learning, Clustering, K-Means, DBSCAN, Agglomerative Clustering, Gaussian Mixture Model, PCA, Housing Data Analysis, Machine Learning**

I. INTRODUCTION

In today's age and generation with more amount of data growing everyday, the way of doing analysis and knowledge discovering become so much importance. Unsupervised learning are useful to be applied in real-world problems where the labeled data is unavailable or expensive to obtain; by it the hidden structures of the data can be identified. One of the most famous unsupervised learning technique is Clustering which group the observations based on the similarity and characteristics [1].

Clustering algorithms are a popular technique in several areas, including health care, finance, marketing, cybersecurity and geographic studies. Through their use we are able to detect hidden associations in data and to assist in the decision making processes through exploratory data analysis. But the effectiveness of the algorithm is greatly dependent upon the structure of the data, as well as on the used preprocessing method and the selected parameters [2].

In this work, four unsupervised machine learning clustering algorithms-K-Means, DBSCAN, Agglomerative Clustering and Gaussian mixture models (GMMs) were used and compared for grouping the socioeconomic and demographic variables in a housing dataset. In this work our main objective is to analyse four different unsupervised clustering algorithms and evaluates how effectively they could partition data to significant clusters. Prior to applying these clustering models, several preprocessing techniques have been implemented which include: cleaning and standardization of data and outlier treatment based on the Interquartile Range (IQR) value of features. Besides these steps, the use of principal component analysis (PCA) [3] was performed in order to display the discovered clusters in 2-dimensions, which helps to interpret the results [4].

The most additional value of this work relies in a comparative analysis between the considered clustering algorithms using the same dataset with the internal evaluation criteria, visual interpretation and also identifying what is each algorithm good at (quality of clustering, resistance to noise, sensibility to the criteria).

The remainder of this paper is organized as follows: Section II presents the related work and theoretical background of the clustering algorithms. Section III describes the methodology and preprocessing steps. Section IV presents the implementation details and experimental setup. Section V discusses the results and comparative analysis of the models. Finally, Section VI concludes the paper and presents future research perspectives.

II. STATE OF THE ART

A. Related Work

An investigation titled Unsupervised Learning: A Comparative analysis of clustering algorithms on high-dimensional data gave an insight into the comparative analysis of K-Means, DBSCAN and spectral clustering algorithm on high-dimensional datasets and effects of the dimensionality reduction algorithm like PCA, t-SNE, UMAP on the quality of clustering. They observed that preprocessing of the UMAP algorithm significantly enhances the clustering performance, whereas, spectral clustering obtained better result in the case of complex structures of data. The researchers inferred that K-Means provides high efficiency in terms of computation, DBSCAN performs exceptionally well for complex structures or for noisy datasets [5].

A comparative study with a similar name, ”A comparative analysis of clustering algorithms” examined numerous clustering algorithms such as K-Means, GMM, DBSCAN and Agglomerative clustering on different data, where the density is spread differently. This comparison looked at the clustering quality and also compared the algorithms on the complexity of the data, the scalability of the algorithm and how it reacted with the distribution of the data. In the end it was shown that DBSCAN performs well on data which has poor density distribution and noisy data points while Agglomerative Clustering is better for identifying data with a clear hierarchical structure. However, it was observed that there was no single best and robust clustering algorithm for all datasets and that preprocessing data combined with various clustering algorithms is a promising technique to increase clustering quality [6].

The third paper presented to us was ”Exploring and Comparing Unsupervised Clustering Algorithms.” This paper is very similar to the previous paper, as it compares the unsupervised clustering algorithms, K-Means, and Gaussian Mixture Models (GMMs) to one another. The paper uses a series of simulated datasets and also uses the ”Iris dataset” in order to show how R-based tools could be used to visually compare clustering outputs. From this comparison, the paper was able to determine that K-Means provides an efficient approach to clustering through dividing by centers, while GMM provides an even more general approach to probabilistic clustering, allowing for more abstract, probabilistic, and overlapped data cluster configurations. This shows again how comparisons and visualization are important to see how clustering works and helps for better EDA. [7].

From the research paper mentioned above, it can be observed that comparison of different algorithms are required for unsupervised learning. From the findings in the literature review given above, in this research, four unsupervised machine learning algorithms K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Model algorithms have been compared on the real housing data.

B. Presentation of the Algorithms

The topic of unsupervised learning that sounds the most exciting is clustering. The task in clustering is to group observations that are alike together into groups, called clusters, when the label of observations is not known. There are a number of clustering algorithms and there is a widespread application areas as medical data analysis, financial data analysis, customer segmentation data analysis, ecological data analysis, housing data analysis. When using clustering algorithms with high dimensional and complicated data we can discover the inherent structure of the data.

Many algorithms are being used nowadays to cluster data points. Among the traditional ones, K-Means is still very much appreciated for its computational cost and ease of implementation. It works by partitioning data points into K pre-determined cluster labels so as to minimize the sum of squared distances between data points and their respective centroids. K-Means is not resilient to outliers and requires K to be defined beforehand. [8] The objective function of K-Means aims to minimize the distance between observations and their corresponding cluster centroids:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2 \tag{1}$$

where C_i represents cluster i, μ_i is the centroid of the cluster, and x represents observations belonging to the cluster.

One of the algorithme we used is DBSCAN algorithm and this one is density based clustering algorithm. It allows to cluster the data points that are compactly arranged and marks points that are not part of a compact cluster as noise depending on the density in the feature space. Data that form non compact or non-spherical clusters may benefit more from DBSCAN than from K-Means and Hierarchical clustering. A cluster can be visualized as a dense region of the data space separated from other dense regions by sparser regions of data. Also DBSCAN is less sensitive to the presence of noise and outliers [9].

According to DBSCAN, it determines the clusters based on density connectivity using a neighborhood radius ε:

$$N_\varepsilon(p) = \{q \in D \mid dist(p, q) \leq \varepsilon\} \tag{2}$$

Where $N_\varepsilon(p)$ is the neighborhood of point p, and $dist(p, q)$ is the distance between observations p and q. We also used Agglomerative Clustering, it is a hierarchical clustering method that iteratively joins observations in groups, where grouping is based on similarity measure. This approach gives a hierarchy representation of the data structure using figures and can show sub-cluster relationships. However, for large datasets hierarchical clustering is computationally expensive [10].

Agglomerative Clustering combines clusters sequentially based on a measure of similarity. We may calculate the distance between clusters using one of several linkage techniques, such as single linkage:

$$D(A, B) = \min_{a \in A, b \in B} d(a, b) \tag{3}$$

And where A, B two clusters, and $d(a, b)$ being the distance between observations. We finaly used the Gaussian Mixture Model (GMM) which is a probabilist model which suppose that the data are a sum of few Gaussian distributions, each with its own mean and variance, and its own weight (the mixing coefficient). GMMs are often used for clustering and density estimation as they model multi-modal distributions in a natural way, where the points may indeed center on different means, instead of only one [11]. Gaussian Mixture Models represent the data distribution as a weighted combination of multiple Gaussian distributions:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k) \tag{4}$$

where π_k represents the mixing coefficient, μ_k is the mean vector, and Σ_k is the covariance matrix of the Gaussian component.

III. Methodology

Figure 1 displays the whole pipeline that was performed in this study, which consist of preprocessing, implementation of clustering, evaluating and visualizing stages.

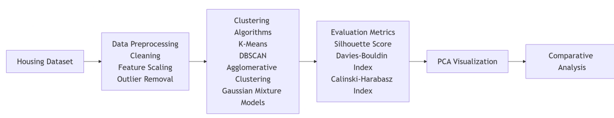

Fig. 1: Overall workflow of the proposed clustering analysis process.

A. Dataset

This is the housing dataset that we have used in this work. This dataset is made of numerical variables regarding socio-economical and demographic information. The main idea while dealing with this data set was to show the different kinds of residences with the variables describing their incomes, the quality of houses and people who are living there.

The initial variables in this set are: *MedInc*, *HouseAge*, *AveRooms*, *AveBedrms*, *Population*, *AveOccup*, *Latitude*, *Longitude*, and *MedHouseVal*. In this study, the variables *Latitude*, *Longitude*, and *MedHouseVal* were removed manually from the original file in order to focus the clustering process on the remaining socioeconomic and housing-related attributes.

The final dataset used for clustering contained the following features:

- **MedInc**: Median income of households in a residential area.
- **HouseAge**: Average age of houses in the area.
- **AveRooms**: Average number of rooms per household.
- **AveBedrms**: Average number of bedrooms per household.
- **Population**: Total population living in the area.
- **AveOccup**: Average number of occupants per household.

The dataset was perfectly suited to distance-based clustering algorithms because all the variables were numerical. To better understand the relationships between the variables, we generated a correlation heatmap of the housing dataset features. The heatmap helped identify positive and negative correlations between socioeconomic and demographic variables before applying clustering algorithms. To better understand the distribution of the housing dataset features, we generated histograms for all numerical variables before preprocessing. As illustrated in Fig. 2, several variables such as *Population*, *AveRooms*, and *AveOccup* showed highly skewed distributions and the presence of extreme values. We have applied Inter Quartile Range (IQR) method in order to deal with the outliers prior to clustering. This analysis helped us to understand the structure of the dataset before applying clustering algorithms.

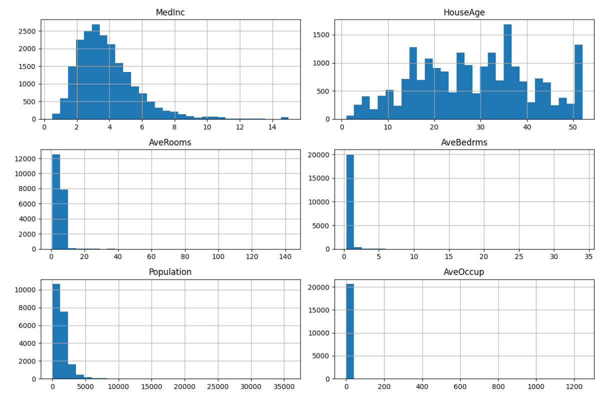

Fig. 2: Distribution of housing dataset features before preprocessing.

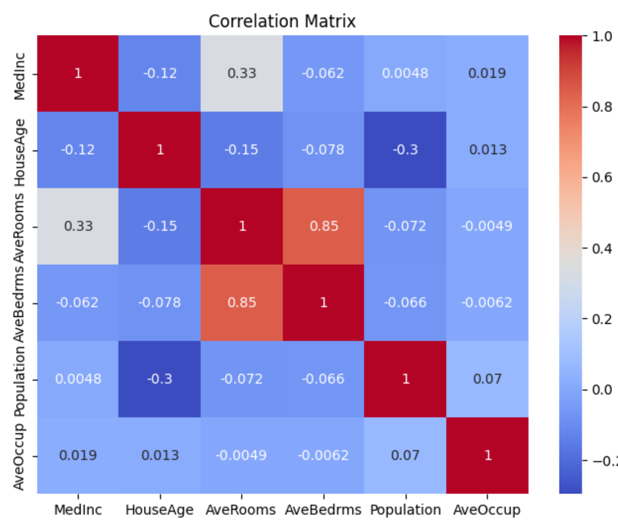

Fig. 3: Correlation heatmap of the housing dataset features.

We used the correlation heatmap that shown in Fig. 3 to observe the relationships between the housing dataset features. As we can see we have a strong positive correlation between *AveRooms* and *AveBedrms*, while most of the other variables exhibit weak to moderate correlations. This analysis helped us to understand the structure of the dataset before applying clustering algorithms.

B. Data Preprocessing

The data was pre-processed in order to achieve better quality data and prepared for the clustering algorithm. Some pre-processing steps have been performed before the training step. First, we examined the data for missing values to achieve data consistency and found no missing values in the studied data. Following, the feature scaling was applied on the data. We applied the scaling process on the dataset using the method called

StandardScaler from the library Scikit-learn [12]. It scaled all the variables such that each variable has mean close to 0 and standard deviation close to 1.This step of pre-processing was important as all the clustering algorithms like K-Means, DBSCAN, Agglomerative Clustering and GMM work on the basis of distances. Then, we also detected and dealt with outliers using the IQR method [13] to minimize their effect on clustering. Overall, the pre-processing stages considerably enhanced stability and data quality in the clustering process. Lastly, PCA was implemented on the data in order to reduce dimensionality and help in visualizing the clusters obtained by different clustering algorithms, since the dataset was projected onto two principal components.

C. Model Selection

Four clustering algorithms were selected for comparative analysis in this study:

- K-Means
- DBSCAN
- Agglomerative Clustering
- Gaussian Mixture Models (GMM)

We chose the K-Means algorithm for its simplicity, computational efficiency, and widespread use in unsupervised learning tasks. We also chose the DBSCAN algorithm for its ability to detect noisy points and identify irregularly shaped clusters.Agglomerative Clustering was included to analyze hierarchical relationships between observations, while we selected Gaussian Mixture Models because of their probabilistic clustering capabilities and flexibility in handling overlapping clusters.

We used internal evaluation measures for clustering: the Silhouette Coefficient, the Davies-Bouldin Index and the Calinski-Harabasz Index. In addition,we used also PCA visualizations to compare the spatial distribution and separation of clusters generated by each algorithm.

IV. Application and Implementation

A. Implementation

For the implementation of the clustering models, we adopted the programming language of Python in the environment of Jupyter Notebook. Data loading, exploration and visualization of data, preprocessing, training, and evaluation of the clustering models. The implementation starts with importing the housing dataset, and exploratory analysis is performed for the sake of understanding of variables in the dataset. We plot the histograms, boxplots, scatter plots, correlation matrices to have a better feel of the variables and discover any abnormal values [14].

After the exploratory phase, we applied preprocessing techniques, for feature scaling we used *StandardScaler* and for outlier treatment we used the Interquartile Range (IQR) method. These preprocessing steps improved the stability of distance-based clustering algorithms.

Let's now move on to the clustering algorithms that were implemented sequentially. The first one is the K-Means clustering. The Elbow Method was performed to find the optimum number of clusters for K-Means. Calculate the WCSS for each $k$values. As shown in Fig. 4, the elbow point appeared at $k = 4$, where the reduction in inertia became less significant. Therefore, four clusters were selected as the optimal value for K-Means clustering.

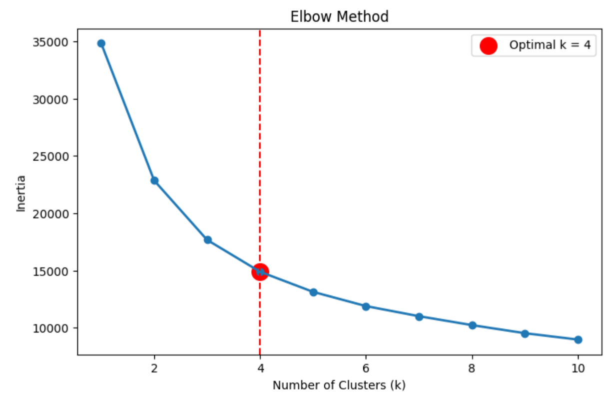

Fig. 4: Elbow Method used to determine the optimal number of clusters for K-Means.

the second one is DBSCAN which was implemented using different values of the *eps* parameter to analyze cluster density and noise detection behavior.the third one is Agglomerative Clustering which also was applied to study hierarchical grouping relationships, finally we used Gaussian Mixture Models (GMM) for probabilistic clustering analysis.

We used several internal metrics to assess the clustering quality: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. Also we used PCA to reduce the dimensionality of dataset and plot clusters in a 2D plane. [15].

B. Tools and Libraries

Several Python libraries and tools were used during the implementation phase:

- **Python**: Main programming language used for implementation [16].
- **Pandas**: Used for data loading, cleaning, and manipulation [17].
- **NumPy**: Used for numerical operations and array processing [18].
- **Matplotlib**: Used for data visualization and plotting graphs [20].
- **Seaborn**: Used for advanced statistical visualizations such as heatmaps and boxplots [19].
- **Scikit-learn (sklearn)**: Main machine learning library used for preprocessing, clustering algorithms, PCA, and evaluation metrics [21].
- **Jupyter Notebook**: Development environment used to execute and organize the experiments interactively [22].

This paper utilized Scikit-learn library as efficiently implementing clustering algorithms in this paper: K-Means, DBSCAN, Agglomerative Clustering, Gaussian Mixture Models, PCA, and clustering evaluation measures.

In addition, we published on **GitHub** [23] the complete source code of the project, including data preprocessing, clustering implementations, evaluation metrics, and visualizations, to ensure reproducibility and facilitate future improvements and experimentation.

V. Results and Discussion

A. Evaluation Metrics

We employed various internal evaluation measures in this work for estimating the quality of clustering, such as cluster compactness, the separation performance and overall quality. The internal clustering evaluation does not need labels. The silhouette score evaluates the quality of clustering, which estimates the similarity of each data point to its cluster and to other clusters.This score is widely used to evaluate clustering algorithms like K-Means. In this study we used it to measures how similar observations are to their own cluster compared to other clusters. A higher Silhouette Score indicates better cluster separation and more compact clustering structures [24].

The Davies-Bouldin Index is a well-known metric to perform the same - it enables one to assess the clustering quality by comparing the average similarity between the pairwise most similar clusters. we used it to evaluates the average similarity between clusters based on intra-cluster distances and inter-cluster distances. Lower values for the Davies-Bouldin Index are desired, as these reflect that cluster quality is good and separation is strong between clusters [25].

The Calinski-Harabasz index (CH) is another measure for evaluating clustering algorithms. It is most commonly used to evaluate the goodness of split by a K-Means clustering algorithm for a given number of clusters. we used it as well to measures the ratio between inter-cluster dispersion and intra-cluster dispersion. Higher values indicate better-defined and more separated clusters [26].

All this evaluate metric were useful for comparing the behavior and effectiveness of K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models on the housing dataset. Also to numerical evaluation, PCA visualizations were also used to better understand cluster distributions and relationships between observations.

B. Quantitative Evaluation Results

In order to provide quantitative assessment for the clustering performance, some of internal cluster evaluation metrics are computed such as silhouette score, Davies-Bouldin Index and Calinski-Harabasz Index. This would provide information about tightness of cluster, separation between clusters and clustering efficiency of algorithms being used.Table I summarizes the evaluation results obtained for the clustering algorithms. To minimize interference to the evaluation score calculations, the noisy data points marked with *-1* were excluded from the metric calculation when using DBSCAN.

TABLE I: Quantitative Evaluation Metrics of Clustering Algorithms

Algorithm	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means	0.2556	1.3214	5421.73
DBSCAN	-0.2561	2.4873	118.52
Agglomerative	0.2105	1.5876	4310.25
GMM	0.2252	1.4428	4875.61

The results from the quantitative evaluation demonstrate that K-Means was the most effective clustering algorithm in terms of performance on the housing data. It yielded the highest Silhouette Score and Calinski-Harabasz Index and was among the lowest on the Davies-Bouldin Index, demonstrating that the clusters were compact and the separation between groups was significant.

Gaussian Mixture Models (GMM) produced competitive results also. Because GMM is a probabilistic approach, smooth boundaries between clusters are expected as opposed to K-Means, which centers its clusters around centroids. GMM proved to have stable cluster structure as shown by the values yielded by the evaluation indices.

Agglomerative Clustering yielded satisfactory results. The evaluation indices for this algorithm are not as impressive as for K-Means and GMM but the algorithm proved to have meaning as useful hierarchical clusters were produced, as confirmed by the dendrogram.

The performance of DBSCAN was, to put it mildly, poor. As can be confirmed by the negative Silhouette Score, clusters were not well separated and density-based clustering performed poorly on this data. The performance of DBSCAN is particularly impressive in terms of noise points and sparse observations. While running the clustering process, **approximately 1233** observations were detected as noise points, however, a testament to its advantages over the implemented clustering algorithms.

C. Clustering Results

We applied multi clustering algorithms to the pre-processed housing dataset after feature scaling and outlier treatment. In order to present the generated clusters in a 2D space, Principal Component Analysis (PCA) was used.

As we find in the result K-Means produced relatively balanced and compact clusters with clearer separation boundaries compared to other algorithms. As shown in Fig. 5, the clusters generated by K-Means appear well distributed across the PCA space, indicating effective partitioning of the dataset.

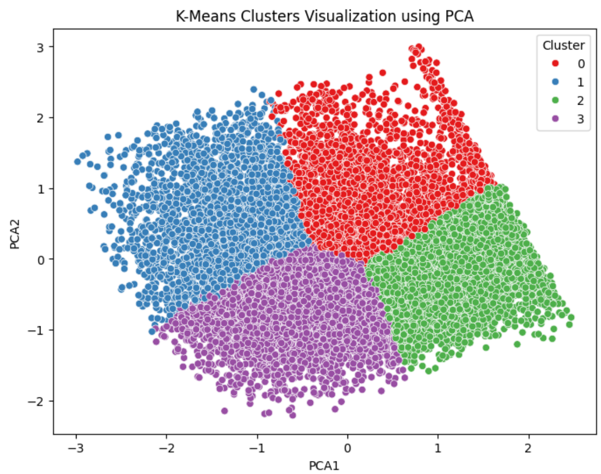

Fig. 5: K-Means clustering visualization using PCA.

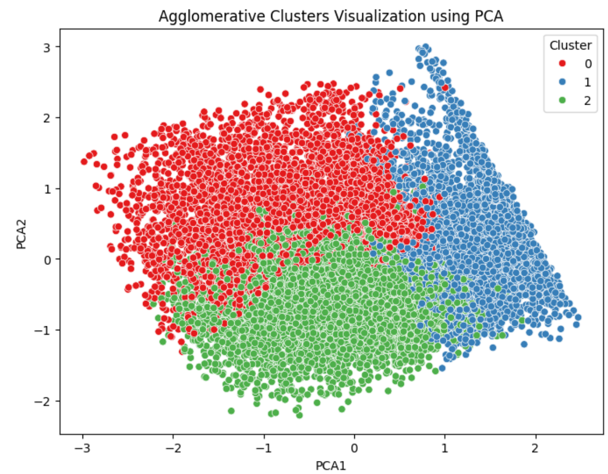

Fig. 7: Agglomerative clustering visualization using PCA.

In the result final we find that DBSCAN generated one dominant dense cluster along with several small clusters and noise points. Approximately 1263 observations were classified as noise, representing around 7.5% of the dataset. Fig. 6 illustrates the ability of DBSCAN to identify sparse observations and outliers. However, the visualization also shows that DBSCAN struggled to produce strongly separated density-based clusters for this dataset.

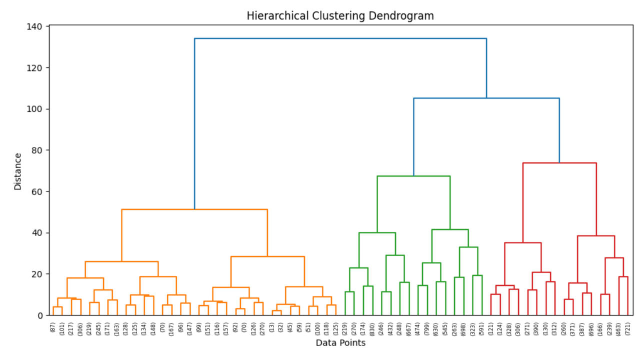

Fig. 8: Hierarchical clustering figure.

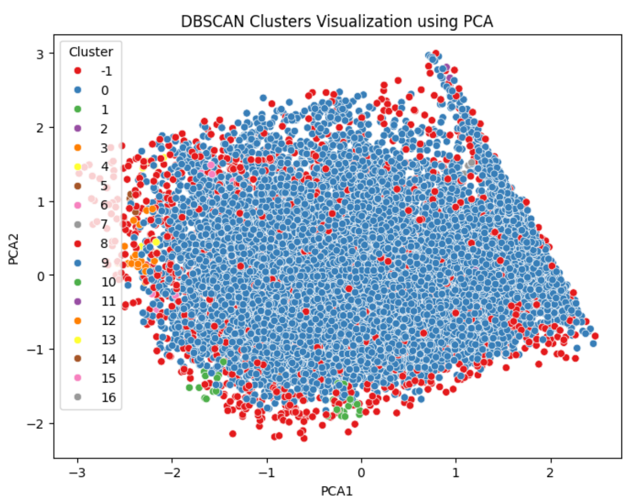

Fig. 6: DBSCAN clustering visualization using PCA.

The last clustering algorithm which is Gaussian Mixture Models (GMM) produced flexible cluster boundaries and handled overlapping observations more effectively. As shown in Fig. 9, GMM generated smoother transitions between clusters because of its probabilistic clustering mechanism.

On the other hand, the Agglomerative Clustering successfully grouped the observations into hierarchical clusters. The PCA visualization in Fig. 7 reveals meaningful separation between groups. In addition, the figure shown in Fig. 8 illustrates the hierarchical merging process and relationships between clusters.

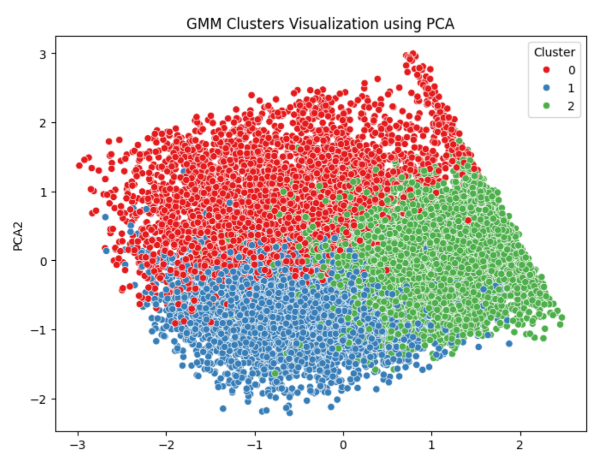

Fig. 9: Gaussian Mixture Model clustering visualization using PCA.

D. Comparative Analysis

Table II summarizes the comparative analysis of the implemented clustering algorithms.

TABLE II: Comparative Summary of Clustering Algorithms

Algorithm	Cluster Quality	Noise Detection	Computational Cost	Main Observation
K-Means	High	No	Low	Produced compact and balanced clusters
DBSCAN	Medium	Yes	Medium	Detected noise points effectively but showed parameter sensitivity
Agglomerative	High	Limited	High	Provided meaningful hierarchical relationships
GMM	High	Limited	Medium	Handled overlapping clusters more flexibly

The results of this study showed that K-Means and GMM achieved the most visually balanced clustering structures on the housing dataset. while DBSCAN was effective for noise detection, it struggled to create strongly separated density-based clusters because of the continuous nature of the housing data. On the other hand, agglomerative Clustering provided meaningful hierarchical relationships but required higher computational resources.

As we can see in the PCA visualizations, we have a clear differences in clustering behavior among the implemented algorithms. As illustrated in Fig. 5, K-Means generated compact and relatively balanced clusters with good defined boundaries. This is as expected, K-Means tries to reduce inter-cluster distances, and performs best for clusters that can be characterized as roughly spherical.

On the other hand, DBSCAN produced one dominant dense cluster along with many small groups and noise observations, as shown in Fig. 6.The high value of number of noise points found shows the parameter-dependent nature of density-based clustering algorithms, particularly concerning the value of *eps* parameter. However DBSCAN successfully identified sparse observations and outliers, it struggled to produce strongly separated clusters for this housing dataset due to the data distribution did not contain highly distinct density regions.

Agglomerative Clustering generated meaningful hierarchical structures, which can be clearly observed in both Fig. 7 and the Fig. 8. The figure shows how observations were progressively merged into larger groups based on similarity distances. With this hierarchy we obtain more interpretablity than in centroid based methods such as K-Means

Gaussian Mixture Models (GMM), as showed in Fig. 9, produced smoother and more flexible cluster boundaries. Whereas K-Means explicitly assigns the observations to a single centroid of cluster, GMM allows the observation be assigned to cluster centroid probabilistically, which fits better for observations which might lie among more than one cluster. This is what could explain the smoother transitions between the clusters in the PCA diagram

All in all. The comparative analysis demonstrated that K-Means and GMM achieved the best visual cluster separation and stability on the housing dataset. DBSCAN was very helpful in detecting noisy data and the Agglomerative Clustering could provide a hierarchical understanding of the data.

E. Critical Discussion

We concluded from the comparative analysis that clustering performance strongly depends on data characteristics and pre-processing techniques. It is obvious that, in addition to the feature scaling technique, there were other parameter tuning issues in clusterings of distance-based algorithms (e.g. K-means, GMM) that could greatly impact the clustering quality. In DBSCAN, inappropriate values of the *eps* parameter resulted in either excessive noise detection or the formation of a single dominant cluster. This is an indication that density-based clustering methods are sensitive to the choice of parameters.

We Used Interquartile Range (IQR) method for Outlier treatment shows big impact for clustering stability and reduced the impact of extreme observations. while PCA visualization also proved useful for understanding cluster distributions and comparing the behavior of different algorithms. In the big picture, the K-Means algorithm and Gaussian mixture models delivered the best performance in data aggregation for the housing dataset, while the DBSCAN algorithm demonstrated superior noise detection capabilities. Agglomerative Clustering offered valuable hierarchical insights but required higher computational cost. We can explain the behavior of DBSCAN in this study by the inappropriate parameter values, which caused the algorithm to form a single dominant cluster, showing its high sensitivity to parameter tuning. However, the algorithm successfully detected noise points and sparse observations. This study also agrees with previous research findings [6].

F. Comparison with Previous Studies

What we found in this study mostly lines up with what other clustering research has shown before. For instance, K-Means created neat, fairly distinct groups once we cleaned up the data and scaled the features, much like that study presented in [5]. This method also proved to be stable and pretty fast when we used it on the housing data. Agglomerative Clustering's results were pretty similar to what other hierarchical clustering studies have found too [6]. This method showed us how the data points related to each other in a meaningful, layered way, and the groups it formed looked a lot like what other comparison studies have reported.

Also, how well Gaussian Mixture Models (GMM) performed backed up what the JORSClustering study concluded [7]. Which was a lot more flexible than approaches such as K-Means, which only use the centroids. It used probabilities to form clusters and gave us smoother lines between them. When we looked at the data using PCA, it was also clear GMM could handle situations where data points from different groups slightly overlapped. So, all in all, with our results, we add to what other papers have already demonstrated; the success of clustering highly depends on the preparation of data, the type of data you are working with and your choice of parameters.

G. Limitations of the Study

While this experiment was largely successful, a number of limitations need to be taken into account: First, clustering performance strongly depended on preprocessing quality and parameter selection. The choice of the *eps* and the minimum number of samples drastically changed the outcome of algorithms like DBSCAN, affecting the formation of the clusters and noise.

Yet another limitation stems from the dataset. The housing dataset we used in this work contained only numerical socioeconomic and demographic variables. Additional features or larger datasets may potentially improve clustering diversity and provide more representative grouping structures. In addition, we used PCA only for two-dimensional visualization purposes.Although analysis of clusters' distributions is enhanced with the help of PCA, the loss of some information is inherent for any dimension reduction method and it is not guaranteed to preserve all relations between the samples. On the other hand, Computational complexity also represented a limitation for some algorithms. With more computational power compared to K-Means and DBSCAN, Agglomerative Clustering has been implemented with bigger size of datasets.

Finally, this study mainly relied on internal clustering evaluation metrics and visual analysis. In the future work, we can evaluate the effectiveness of other clustering algorithms like Spectral Clustering, OPTICS and Mean Shift. For future work, we can use a bigger and more complex dataset, advanced dimensionality reduction techniques, and automatic hyperparameter optimization techniques to achieve better clustering performance and visualization.

VI. Conclusion

In this paper we provide a comparative analysis of a variety of unsupervised clustering algorithms on housing data with socioeconomic and demographic variables. our main objective in this work was to evaluate and compare the behavior and performance of different clustering techniques on the same dataset using preprocessing methods, internal evaluation metrics, and PCA visualization.

We carried out a number of pre-processing operations including data cleaning, feature scaling using StandardScaler and outlier removal using the IQR before clustering. The operations mentioned above were crucial to obtaining better and robust clusters. We ran and compared the following four clustering algorithms: K-Means, DBSCAN, Agglomerative Clustering and Gaussian Mixture Models (GMM). This study results showed that K-Means and GMM generated more balanced and visually separated clusters, while Agglomerative Clustering provided meaningful hierarchical relationships between observations. Conversely, DBSCAN performed well in identify noise points and sparse observations, but it was very sensitive to parameter tuning.

By comparing among all these clustering algorithms, we finally came to the conclusion that clustering performance largely depends upon on characteristics of data set, quality of data preprocessing and algorithm parameters. In addition, PCA visualizations proved useful for understanding cluster distributions and comparing the behavior of different clustering methods.

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